

MLFF Architecture Revision 84

CUEQ-REPEAT1-DIAG2 full self-tail and deterministic-control refinement

mdstats project

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Revision 84

Release: mdstats 0.20.217a0

Dependency-graph schema: 66

Gate: CUEQ-REPEAT1-DIAG2

The MPA-0/default 10-repeat workstation evidence showed e3nn-self **Fmax** max **2.120e-5**, CuEq-self max **2.076e-5**, and paired e3nn/CuEq max **1.699e-5**, with paired **Frmse** only **2.023e-6** to **3.374e-6** and identical selection in 10/10 repeats. Same-backend noise therefore exceeds the observed cross-backend extreme-value discrepancy.

Revision 84 keeps the active TRAIN2 FP32 parity policy unchanged at **rtol=1e-5**, **atol=1e-5** and refines measurement only. Ordinary doctor output now prints e3nn-self and CuEq-self **Fmax**, **Frmse**, **Fp99**, **Fp99.9**, and counts above **1e-5**, in addition to the existing paired cross-backend statistics.

A second isolated deterministic-control subprocess is added. Before CUDA initialization it sets the CUBLAS deterministic workspace to **:4096:8**, enables deterministic PyTorch algorithms with error-mode diagnostics, disables cuDNN benchmarking, and enables deterministic cuDNN execution. It repeats the same 10-run probe. Successful controls print a complete **[DIAG-DET]** block; unsupported nondeterministic operations print and persist their exact failure without fallback.

Both ordinary and deterministic-control measurements remain non-authorizing. The revision-82 FINAL-GPU1/HF2 handoff remains archival until the measured distributions are interpreted and a final noise-normalized parity criterion is frozen.